JEE

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Maths

Quadratic Equations

DPP: 4

Q1 If α, β are the roots of $x^2-3x+1=0$ then the equation whose roots are $\left(\frac{1}{\alpha-2}, \frac{1}{\beta-2}\right)$ is

(A) $x^2+x-1=0$

(B) $x^2+x+1=0$

(C) $x^2-x-1=0$

(D) None of these

Q2 If α, β are the roots of the equation $ax^2+bx+c=0$, then equation whose roots are $\alpha+\frac{1}{\beta}$ and $\beta+\frac{1}{\alpha}$ is

(A) $acx^2+(a+c)bx+(a+c)^2=0$

(B) $abx^2+(a+c)bx+(a+c)^2=0$

(C) $acx^2+(a+b)cx+(a+c)^2=0$

(D) $acx^3+(a+c)bx+(a+c)^3=0$

Q3 If α, β are the roots of the equation $x^2+px+q=0$, then $-\frac{1}{\alpha}, -\frac{1}{\beta}$ are the roots of the equation

(A) $x^2-px+q=0$

(B) $x^2+px+q=0$

(C) $qx^2+px+1=0$

(D) $qx^2-px+1=0$

Q4 If α, β, γ are the roots of the equation $x^3+4x+1 \qquad =0,$ then $(\alpha+\beta)^{-1}+(\beta+\gamma)^{-1}+(\gamma+\alpha)^{-1}$ is

(A) 2 (B) 3

(C) 4 (D) 5

Q5 Let α and β be the roots of $x^2-6x-2=0$, with $\alpha > \beta$. If $a_n=\alpha^n-\beta^n$ for $n \geq 1$, then the value of $\left(a_{10}-2a_8\right)/2a_9$ is

(A) 1 (B) 2

(C) 3 (D) 4

Q6 Let p and q be two real numbers such that $p+q=3$ and $p^4+q^4=369$. Then $\left(\frac{1}{p}+\frac{1}{q}\right)^{-2}$ is equal to

Q7 The sum of all the real roots of the equation $\left(e^{2x}-4\right)\left(6e^{2x}-5e^x+1\right)=0$ is

(A) $\log_e 3$

(B) $-\log_e 3$

(C) $\log_e 6$

(D) $-\log_e 6$

Q8 If α and β be two roots of the equation $x^2-64x+256=0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}}+\left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}}$ is

(A) 1 (B) 3

(C) 2 (D) 4

Q9 The sum of all real values of x satisfying the Equation $\left(x^2-5x+5\right)^{x^2+4x-60}=1$ is:-

(A) 5 (B) 3

(C) -4 (D) 6

Q10 Solve the equation:

$$(6-x)^4+(8-x)^4=16$$

Q11 The roots of the equation $4x^4-24x^3+57x^2+18x-45=0$. If one of them is $3+i\sqrt{6}$, are

(A) $3-i\sqrt{6}, \pm\sqrt{\frac{3}{2}}$

(B) $3-i\sqrt{6}, \pm\frac{3}{\sqrt{2}}$
(C) $3-i\sqrt{6}, \pm\frac{\sqrt{3}}{2}$
(D) None of these

Q12 If the roots of the equation $x^3-12x^2+39x-28=0$ are in AP, then their common difference will be
(A) ±1 (B) ±2
(C) ±3 (D) ±4

Q13 Let α, β and γ are the roots of the cubic $x^3-3x^2+1=0$. Find a cubic whose roots are $\frac{\alpha}{\alpha-2}, \frac{\beta}{\beta-2}$ and $\frac{\gamma}{\gamma-2}$. Hence or otherwise find the value of $(\alpha-2)(\beta-2)(\gamma-2)$

Q14 If α, β are roots of $ax^2+bx+c=0$, then the equation $ax^2-bx(x-1)+c(x-1)^2=0$ has roots
(A) $\frac{\alpha}{1-\alpha}, \frac{\beta}{1-\beta}$
(B) $\frac{1-\alpha}{\alpha}, \frac{1-\beta}{\beta}$
(C) $\frac{\alpha}{1+\alpha}, \frac{\beta}{1+\beta}$
(D) $\frac{1+\alpha}{\alpha}, \frac{1+\beta}{\beta}$

Q15 If α and β are the roots of the equation $x^2+px+2=0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation $2x^2+\ 2qx+1=0$, then $\left(\alpha-\frac{1}{\alpha}\right)\left(\beta-\frac{1}{\beta}\right)\left(\alpha+\frac{1}{\beta}\right)\left(\beta+\frac{1}{\alpha}\right)$ is equal to
(A) $\frac{9}{4}\left(9-q^2\right)$
(B) $\frac{9}{4}\left(9+p^2\right)$
(C) $\frac{9}{4}\left(9+q^2\right)$
(D) $\frac{9}{4}\left(9-p^2\right)$

Q16 If $\alpha, \beta \,\&\, \gamma$ are the roots of the equation $x^3-x-1=0$ then, $\frac{1+\alpha}{1-\alpha}+\frac{1+\beta}{1-\beta}+\frac{1+\gamma}{1-\gamma}$ has the value equal to:
(A) zero (B) -1
(C) -7 (D) 1

Q17 If α, β and γ are the roots of equation $x^3-3x^2+x+5=0$ then $y=\sum\alpha^2+\alpha\beta\gamma$ satisfies the equation
(A) $y^3+y+2=0$
(B) $y^3-y^2-y-2=0$
(C) $y^3+3y^2-y-3=0$
(D) $y^3+4y^2+5y+20=0$

Answer Key

Q1	(C)
Q2	(A)
Q3	(D)
Q4	(C)
Q5	(C)
Q6	4
Q7	(B)
Q8	(C)
Q9	(B)
Q10	6 or 8
Q11	(C)
Q12	(C)
Q13	$3x^3 - 9x^2 - 3x + 1 = 0$ and 3
Q14	(C)
Q15	(D)
Q16	(C)
Q17	(B)